

Statistics of Measurement (Lectures 1-4)

Lecture 1 - Introduction and properties of probabilities

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1 Introduction

Statistics is a study of two somewhat separate issues, which we shall call **probability** and **estimation** (although people will often use ‘statistics’ to mean just the latter). For physicists, the main aim of studying statistics is to be able to interpret the results of experiments so as to derive information on the values and uncertainties of physical quantities. This is essentially the ‘estimation’ part; we estimate values, e.g. for the mass of the electron, using data we acquire during an experiment.

However, no experiment can measure a quantity *exactly*, so there will inevitably be some level of random error associated with each value obtained. Hence, estimation requires a knowledge of ‘probability’, and so in essence, we must study this aspect first. We can think of the study of probability as being the way of predicting how often we would get the various possible outcomes of our experiment, e.g. how often we would measure the electron mass to have some particular value. Hence, you can think of the two as going in opposite directions; i.e. studying probability allows one to take physical constants and predict the likely outcomes of an experiment, whilst estimation takes the actual outcome of a particular experiment and finds the values of physical constants which are compatible with the data. You will sometimes hear estimation referred to as the ‘inverse problem’ to probability.

One thing to note however is that ‘probability’ and ‘estimation’ are two very different topics: Probability is a mathematically rigorous field; whilst, estimation is less straightforward, often with no absolutely ‘right’ answer and is to some extent arbitrary. As we will see later, there is even a basic disagreement at the fundamental level as to what we should be estimating. It is common for students to be fairly comfortable with probability but can be confused with estimation; I hope the latter will not be true for you. We will cover probability in the first part of the lecture course and estimation in the second part. We will not deal with systematic errors greatly, but you should be aware of these, partly through your experience in labs this year.

2 Frequentist and Bayesian probabilities

Everyone will have an intuitive feel for what we mean by a probability; e.g. “The probability of throwing a die and getting a six is $1/6$ ”. However, it turns out that actually defining probability can be quite tricky.

The most obvious definition would be that the probability of something is the fraction of times it occurs when the number of ‘experiments’, sometimes called ‘trials’, becomes very large. In fact the actual definition would be the limit of this fraction as the number of experiments tends toward infinity. This is called the **frequentist** (or ‘classical’) definition, as it defines probability through the frequency with which something occurs in a large number of experiments. This definition means that the probability will be a fixed value for a given experimental setup and everyone will use the same value. Note, the critical concept here is *repeatability*.

In contrast, the **Bayesian** idea of probability (named after Thomas Bayes) is that it is related to the *degree of belief* in the outcome, which will depend on the available knowledge of the experimenter. Depending on the extent of their knowledge, different people might therefore assign different probabilities to the outcome for the same experiment. The positive side is that this gets around an issue with the frequentist definition, which is that it is often difficult or impossible to repeat the experiment. The most extreme example would be cosmology studies

of the development of the Universe, which can predict e.g. the probability distribution for sizes of galaxies. We cannot redo the ‘experiment’ of the creation of the Universe to check our probabilities and it is not clear if a frequentist probability definition is even conceptually sensible. The Bayesian approach can be seen as generalising the frequentist definition. However, the downside is that it does mean that probabilities can be harder to evaluate uniquely in the Bayesian approach. For most of the estimation part of the course we will take the frequentist view, while in the last lecture and the second seminar we will discuss the Bayesian approach explicitly.

3 General properties

Although the definition of probability is tricky, the mathematical properties of probabilities are uncontentious and well-defined. If the possible outcomes of an ‘experiment’ (in the general sense, e.g. throwing a die) are x_i with probabilities $P(x_i) = P_i$, then the P_i have to satisfy the **axioms of probability** introduced by Andrey Kolmogorov in 1933. The probability of an event is a non-negative real number, the probability of x_i or x_j is the sum of the probabilities of i and j , and the sum of probabilities of all possible outcomes is 1:

$$P_i \geq 0, \quad P_{i \text{ or } j} = P_i + P_j, \quad \sum_i P_i = 1$$

Many other properties can be derived from these; e.g. clearly every $P_i \leq 1$, so probabilities are bounded between 0 and 1. Also, the probability of something not happening must be

$$P_{\text{not } i} = \overline{P_i} = 1 - P_i = \sum_{j \neq i} P_j.$$

4 Random variables and probability distributions

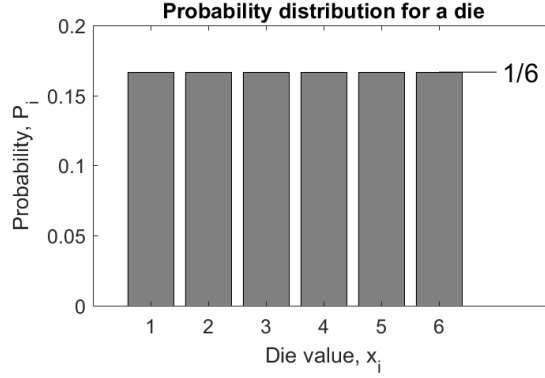
Formally, for any given ‘experiment’, the possible outcomes are called the ‘sample space’ and each outcome will have a probability assigned to it. e.g. if I throw a fair die, the sample space is the set of values $\{1, 2, 3, 4, 5, 6\}$ and by symmetry, each has a probability of $1/6$ so that they sum to one as required. If I throw a fair coin, the sample space is $\{\text{Heads}, \text{Tails}\}$ and again by symmetry, each must have a probability of $1/2$ so as to sum to one. The quantities which we measure, and which can take any one of the values in the sample space, are called ‘random variables’; in these examples they are the top face of the die or coin. Each time we do an experiment, the outcome is randomly picked from one of the outcomes in the sample space, with an appropriate probability assigned to it.

In both these cases, the probabilities P_i for each of the outcomes within the sample space x_i are equal, by assumption. The probability distribution is **uniform**. For a uniform distribution with N possible outcomes, let $P_i = k$, a constant. We can use the normalisation of all the probabilities to fix the constant; since

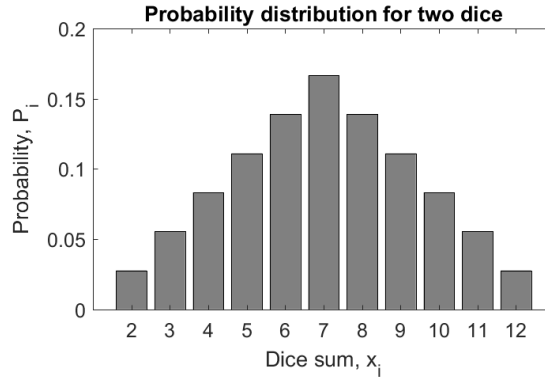
$$\sum_{i=1}^N P_i = Nk = 1 \quad \text{then} \quad P_i = k = \frac{1}{N}$$

which agrees with the probabilities we assigned above, with $N = 6$ for the die and $N = 2$ for the coin. We can draw such a probability distribution for all elements in the sample space as a graph; e.g. the case of the die is shown below.

In general, probabilities are NOT typically all equal; for example, the sum of two dice is not uniform, even though the outcome of each die independently is. This is straightforward to



understand; any given outcome of the first die and the second die has a probability of $(1/6)^2 = 1/36$. There is only one way to make the sum be 2 (or 12), namely two ones (or two sixes), and so these indeed each have a probability of $1/36$. However, there are two ways to make a sum of 3, namely a one on the first die and a two on the second, or vice versa. Therefore, a sum of 3 has a probability equal to the sum of the separate probabilities (using Kolmogorov's axioms) and hence is $1/36 + 1/36 = 1/18$. The same holds for 11. The probabilities increase for larger sums than 3, as there are more and more combinations which give a particular sum, and peak at the sum of 7, which has six combinations and so a probability of $6/36 = 1/6$. The probability distribution for this is shown below. Note here, as in all probability distributions, the total area under the distribution must sum to one to satisfy Kolmogorov's axioms.



5 Expectation value and variance

The **expectation value**, or 'expected' value, is the long-run average value of the distribution.

You can think of it as the average that would be obtained from a very large number of experiments, given the distribution. Since the probability of getting a value x_i is P_i , then for $N \rightarrow \infty$ experiments, the number of outcomes equal to x_i would be NP_i (from the frequentist definition of probability). The expectation value is then given by

$$E(x) = \langle x \rangle = \frac{1}{N} \sum_i x_i (NP_i) = \sum_i x_i P_i$$

For the case of two dice, this gives

$$E(x) = \langle x \rangle = \frac{1}{36}(2 + 12) + \frac{2}{36}(3 + 11) + \frac{3}{36}(4 + 10) + \frac{4}{36}(5 + 9) + \frac{5}{36}(6 + 8) + \frac{6}{36}(7) = 7$$

as is obvious from the symmetry of the probability distribution. For any uniform distribution, the expectation value is

$$E(x) = \langle x \rangle = \sum_i x_i \frac{1}{N} = \frac{1}{N} \sum_i x_i$$

i.e. what we think of as the usual average. For a single die, then this would be

$$E(x) = \langle x \rangle = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = \frac{7}{2} = 3.5$$

This shows that the expectation value does *not* have to be one of the possible outcomes from the probability distribution in general; it happens to be so for two dice but not for one die.

It is often useful to have a measure of the spread of values in a distribution. The **variance** is the expectation value of the square of the difference of the x_i from the average value

$$V(x) = \langle (x - \langle x \rangle)^2 \rangle$$

This can be written in a more useful form

$$V(x) = \langle x^2 - 2x\langle x \rangle + \langle x \rangle^2 \rangle = \langle x^2 \rangle - 2\langle x \rangle \langle x \rangle + \langle x \rangle^2 \langle 1 \rangle = \langle x^2 \rangle - \langle x \rangle^2$$

which means

$$V(x) = \left(\sum_i x_i^2 P_i \right) - \left(\sum_i x_i P_i \right)^2$$

Variance is commonly used rather than just the average deviation from the mean because positive and negative deviations will cancel to get an average deviation of zero. Also, as you have used in labs, the variance of the sum of two independent distributions is the sum of their variances, but the same is not true for the deviation from the mean. To get a quantity related to the spread which has dimensions of $[x]$, we take the square root of the variance to calculate the **standard deviation**.

For a die, the variance is

$$V(x) = \frac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) - \left(\frac{7}{2} \right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} = 2.92$$

and so the standard deviation is $\sqrt{2.92} = 1.71$. This gives some idea of the spread of the distribution.

6 Two dimensional probability distributions

So far we have discussed probability distributions of x_i in one dimension. It is possible to generalise the idea into more than one dimension, e.g. for two dimensions, where we would have x and y random variables related to two different sets of outcomes.

Let's take a simple example of rolling a die and also tossing a coin, so x is the die outcome and y the coin outcome. The x_i all have probability $1/6$ and the y_i have $1/2$. We define the **joint probability** for the outcomes x_i and y_j as $P_{ij} = P(x_i, y_j)$. If I roll the die and toss the coin, and ask for the probability of getting a six on the die as well as getting heads on the coin, then the probabilities are multiplied

$$P(\text{Six, Heads}) = P(\text{Six}) \times P(\text{Heads}) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

However, as we will see, this is true only because they are **independent**. The probability of the die outcome is unrelated to the coin outcome so these two are indeed independent. It is often convenient to display these values as a table, showing the probabilities of the possible outcomes we are interested in, as shown below, where for simplicity we only care if we get a six or not.

		Heads	Tails
		1/2	1/2
Six	1/6	1/12	1/12
Not-six	5/6	5/12	5/12

The table above is a representation of the two-dimensional probability distribution. Each entry in the main part of the table corresponds to one of the joint probabilities; for example $P(\text{Six, Heads})$ corresponds to the upper left entry. In this case, the probability in each entry in the table is the product of the two overall probabilities, again only because they are independent. The columns sum to the probabilities written along the top, which give the probabilities for the coin outcome by itself. Similarly, the rows sum to those on the left side, which are the die outcome probabilities. [Nb. Strictly speaking, the case of two dice discussed above is another independent 2D case, but by only being interested in the sum of the dice values, we reduced it to a 1D distribution.]

In general, the probabilities in a 2D distribution do not have to be independent; if not, they are said to be dependent or **correlated**. Here's an example that should illustrate the point: within the UK population, 50% of people are female and 50% are male. Also, approximately 1% of people are pregnant and 99% are not pregnant. However, the probability of being male and pregnant is not

$$P(\text{Male, Pregnant}) = P(\text{Male}) \times P(\text{Pregnant}) = 0.5\%$$

but is effectively zero. The equivalent table in this case is shown below.

		Male	Female
		50%	50%
Pregnant	1%	0%	1%
Not pregnant	99%	50%	49%

Note the entries, i.e. the joint probabilities, are *not* the products of the overall probabilities in this case. Hence, in general the joint probabilities only factorise to $P(x_i, y_j) = P(x_i) \times P(y_j)$ in the special case of independent variables x and y . In the general case, having dependent variables means that if we know x_i , then the probability for y_j is different from what it would be if we knew nothing about x , i.e. $P(y_j)$ by itself. We write this as $P(y_j|x_i)$, the **conditional probability**, which should be read as the probability of getting y_j , given that x_i is known. Similarly, if we know y_j , then the probability of getting x_i is changed from $P(x_i)$ to $P(x_i|y_j)$. For the special case of independent variables, then $P(x_i|y_j) = P(x_i)$ (as it is not dependent on y_j) and similarly $P(y_j|x_i) = P(y_j)$.

Since the joint probability of x_i and y_j happening together must be given by the probability of x_i and then the conditional probability of y_j given x_i , or the probability of y_j and then the conditional probability of x_i given y_j , then we have

$$P(x_i, y_j) = P(x_i)P(y_j|x_i) = P(y_j)P(x_i|y_j) \quad (1)$$

We can find the conditional probabilities from the joint probabilities using Equation 1 above; for example

$$P(y_j|x_i) = \frac{P(x_i, y_j)}{P(x_i)}$$

The conditional probability is a probability distribution in the first variable only, not the second (as that is known). Hence, the normalisation of the probability distribution requires for any x_i

$$\sum_j P(y_j|x_i) = 1$$

Be careful when using conditional probabilities as they are *not* equal to each other. For the example above, having selected a person who is female, the probability of the person also being pregnant is

$$P(\text{Pregnant}|\text{Female}) = \frac{P(\text{Female, Pregnant})}{P(\text{Female})} = \frac{0.01}{0.50} = 2\%$$

while having selected a person who is pregnant, the probability of them also being female is

$$P(\text{Female}|\text{Pregnant}) = \frac{P(\text{Female, Pregnant})}{P(\text{Pregnant})} = \frac{0.01}{0.01} = 100\%$$

as you would expect.

7 Bayes' theorem

The equality of the last two terms in equation 1 is called **Bayes' Theorem** (after the same Thomas Bayes for which the probability definition above is named). This allows us to convert between the two conditional probabilities, i.e.

$$P(y_j|x_i) = \frac{P(x_i|y_j)P(y_j)}{P(x_i)}$$

Essentially, we can use Bayes' theorem to improve an estimate of the probability of y_j given new information about the probability of x_i and the conditional probability of x_i given y_j . In the example above, if we want to know the probability of a person being pregnant without any other information, our first guess would be the probability of any person being pregnant, 1%. Then, if we received new information that the person was also female, we could use Bayes' Theorem to update the probability of that person being pregnant, given that the person was female

$$P(\text{Pregnant}|\text{Female}) = \frac{P(\text{Female}|\text{Pregnant})P(\text{Pregnant})}{P(\text{Female})} = \frac{1 * 0.01}{0.50} = 2\%$$

We thus scale the initial, or prior, probability with the new information to produce an improved posterior estimate.

We do need to be careful with the denominator here. We can combine the rules above to give the 'marginalisation rule'. The total probability for any x_i must be the sum of the joint probabilities of x_i with every possible outcome of y , i.e.

$$P(x_i) = \sum_j P(x_i, y_j) = \sum_j P(x_i|y_j)P(y_j)$$

This is often the best way to calculate the denominator when using Bayes' theorem. Bayes' theorem itself then reads

$$P(y_j|x_i) = \frac{P(x_i|y_j)P(y_j)}{\sum_{j'} P(x_i|y_{j'})P(y_{j'})}$$

Hence the denominator effectively acts as a normalisation for the terms in the numerator so that

$$\sum_j P(y_j|x_i) = 1$$

for any x_i , as required.

Bayes' theorem is incredibly useful and it has applications today in the modelling of climate change, astrophysics and stock market analysis to name but a few examples. Bayes' Theorem can also be used for probability distributions and in cases where the prior or new information is not very precise but still useful in constraining a parameter of interest. We will return to Bayesian estimation towards the end of this course.

Statistics of Measurement

Lecture 2 - Discrete probability distributions

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8 Introduction

We have seen a few probability distributions in the last lecture, namely those for tossing a coin, and throwing one or two dice. These are examples of discrete (as opposed to continuous) distributions, meaning the outcomes can only have specific values, e.g. ‘Heads’ or ‘Tails’ for the coin, or integers for the dice. We will look at two important discrete distributions for physics in some detail in this lecture and will discuss continuous distributions in the next lecture. The two distributions considered today are the **binomial** and the **Poisson** distributions.

9 Binomial distribution

In the last lecture, we considered tossing a coin to get heads or tails. What if we toss it 10 times; what is the probability of getting e.g. 3 heads? For a process with just **two possible outcomes**, that is independently repeated N times (or ‘trials’), the binomial distribution tells us the probability of getting n of one of the two outcomes. This clearly applies to tossing a coin, where there are obviously only two outcomes. However, it seems not to apply to a die, where there are six outcomes. But this is a matter of what we are interested in; if we restrict ourselves to asking if the die gives a six or not-six, then we have reduced the situation to two outcomes and so can use the binomial for this too.

Let’s take the six or not-six example and see how it works. Take a simple case; how many times will we get two sixes with three throws of a die? Clearly, with one throw of the die, we will get six with probability $1/6$ and not-six with probability $5/6$. Let $p = 1/6$ so $5/6 = 1 - p$. (Note, $1 - p$ is often written as q in some textbooks.) We could get a six on the first throw (with probability p) and again on the second throw (again with probability p) and then a not-six on the last throw (with probability $1 - p$), giving a total probability of $p \times p \times (1 - p) = p^2(1 - p)$ as they are independent. However, there are clearly other ways to get two sixes. One is to get a six, a not-six and then a six, with probability $p \times (1 - p) \times p = p^2(1 - p)$ and the other is to get not-six, six, six with probability $(1 - p) \times p \times p = p^2(1 - p)$. Hence, each of these combinations with two sixes and one not-six has the same probability. If we are not interested in the order in which the sixes appear, but only in the total number of sixes, then (by Kolmogorov) we add the probabilities for all combinations of getting two sixes, which here gives $3p^2(1 - p)$.

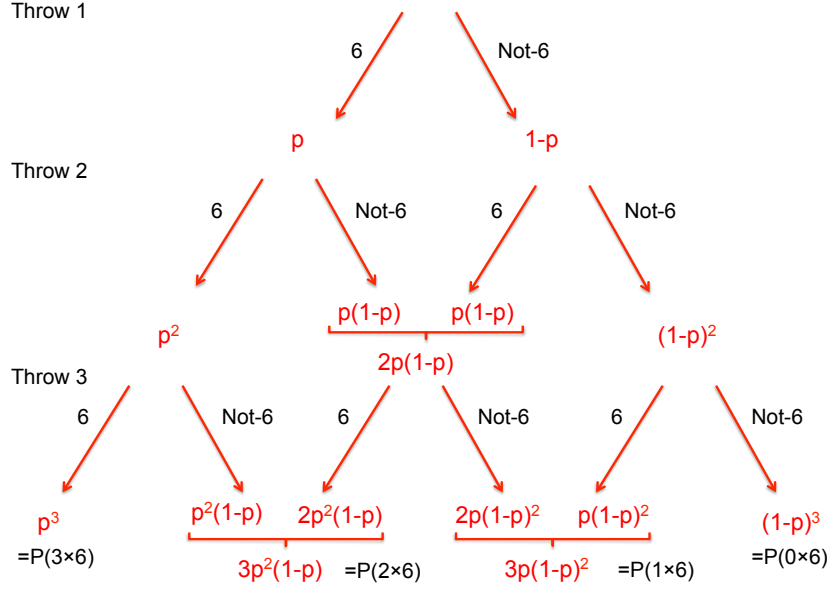
In general, for N throws, then to get n sixes, and hence $N - n$ not-sixes, the probability of any particular combination is $p^n(1 - p)^{N-n}$. The numbers of combinations is given by the ‘binomial coefficient’

$$\binom{N}{n} = \frac{N!}{n!(N - n)!}$$

so-called as they are the values in **Pascal’s triangle**, which are the coefficients which appear in a binomial expansion

$$(a + b)^N = \sum_{n=0}^N \binom{N}{n} a^n b^{N-n} = \sum_{n=0}^N \frac{N!}{n!(N - n)!} a^n b^{N-n}$$

The relationship between these two is made clearer in the figure, showing the various possible outcomes of three dice throws.



Hence, the probability of getting n sixes out of N throws is given by

$$B(n; p, N) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n}$$

where $0!$ is defined to be 1. For $n = 2$ and $N = 3$, we get

$$B(2; p, 3) = \frac{3!}{2!1!} p^2 (1-p) = 3p^2 (1-p)$$

as before. For $p = 1/6$, this is numerically $= 0.069$.

Note the notation here; the random variable appears first and the parameters of the distribution appear after a semi-colon. The above expression is called the **binomial distribution**, named because of its close relation to the binomial expansion. It holds for any case of two (hence ‘bi’ in ‘binomial’) possible outcomes per independent trial. For example, a given nucleus will decay in some time period with a probability p ; there are two outcomes at the end of the time period; it has decayed or not. If we have N such nuclei, then each is an independent ‘trial’ and the total number that have decayed will be given by the binomial distribution.

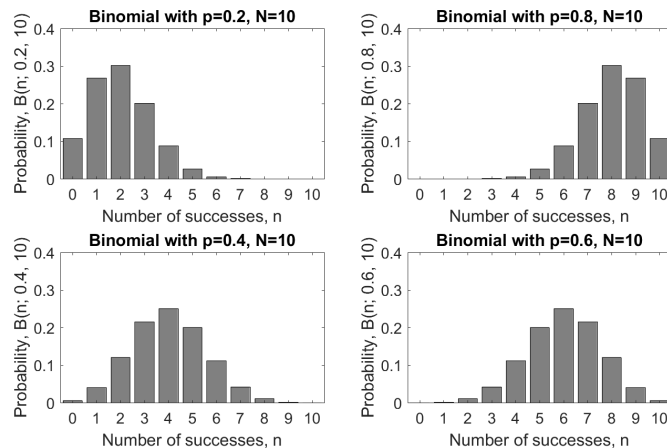
Another example: the probability of getting heads 3 times when tossing a fair coin 10 times is

$$B(3; 0.5, 10) = \frac{10!}{3!7!} 0.5^3 0.5^7 = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} (0.5)^{10} = 0.117$$

Some binomial distributions for varying p and keeping N fixed at 10 are shown below. Clearly n can only have values between 0 and N . Basically, the outcome of each trial is one of two categories and the number of outcomes of one of these categories has the probability given by the binomial distribution. However, it should be clear that the other category will also follow a binomial as they are both possible outcomes. Specifically, the probability of the other is clearly $p' = 1 - p$ so $p = 1 - p'$, and the number of times the other occurs will be $n' = N - n$ so $n = N - n'$, such that

$$B(n; p, N) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} = \frac{N!}{(N-n')!n'!} (1-p')^{N-n'} p'^{n'} = B(n'; p', N)$$

Hence, **the binomial is completely symmetric** with regard to which of the two outcomes is considered. This symmetry can be seen in the plots; the plots for $p = 0.2$ and 0.8 are mirror images of each other, as are $p = 0.4$ and 0.6 . Note the particular outcome being considered is often described as a *success* but the other outcome, the *failure*, also follows the binomial distribution.



The relationship to the binomial expansion makes it easy to see that the probabilities sum to one. The binomial expansion states

$$(a + b)^N = \sum_{n=0}^N \frac{N!}{n!(N-n)!} a^n b^{N-n}$$

With $a = p$ and $b = 1 - p$, this reads

$$[p + (1 - p)]^N = 1^N = 1 = \sum_{n=0}^N \frac{N!}{n!(N-n)!} p^n (1 - p)^{N-n} = \sum_{n=0}^N B(n; p, N)$$

Since the right-hand side is the sum over all the binomial distribution probabilities, the distribution is indeed normalised as required.

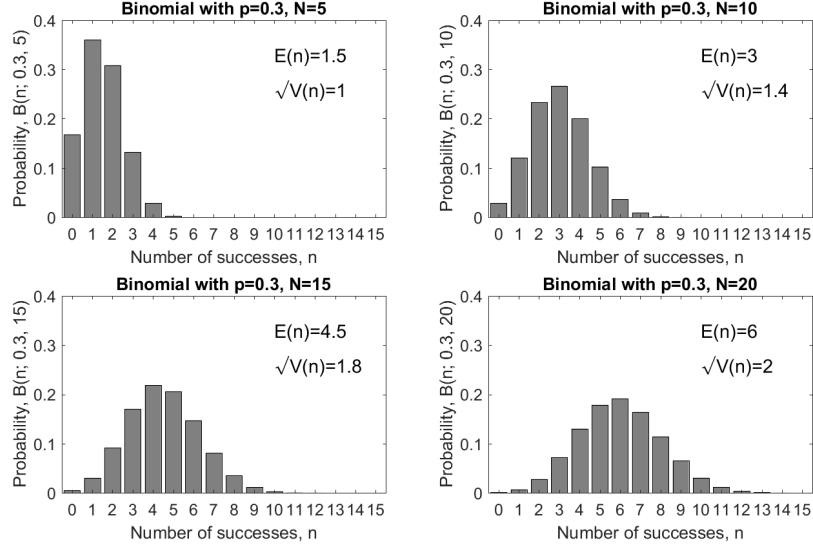
We defined the expectation value and variance of the outcomes for a probability distribution in the last lecture. These can be evaluated for a binomial (see the appendix) and are

$$E(n) = pN, \quad V(n) = Np(1 - p)$$

and hence the standard deviation is $\sqrt{Np(1 - p)}$. The expectation value is the intuitive result; if there is a probability per trial of p , then the average number we would expect in N trials is pN . Plots of binomial distributions with the expectation values and standard deviations for p fixed at 0.3 but varying N are shown below. It can be seen how the expectation values and standard deviations increase with N . As you would expect, the number of successes increases with the number of trials, and the probability distribution spreads out as there are more potential combinations of outcomes.

10 Non-examinable: multinomial distributions

If we want to know the probability of throwing a die and getting n_6 sixes and n_5 fives out of N throws, then we need to go beyond the binomial distribution. We can actually find the



probability for this using the binomial twice. Firstly, it gives us the probability for n_6 sixes and therefore $N - n_6$ not-sixes, as before. However, we can then consider the not-sixes and use another binomial to give the probability of n_5 fives out of the $N - n_6$ not-sixes. The total probability is the product of the two terms. Since there are three possible outcomes (six, five and not-six-or-five) then the resulting distribution is called a trinomial, rather than a binomial. You will probably not be surprised to know that the coefficients in the trinomial distribution are the same as those that arise in the expansion of $(a + b + c)^N$. Also, this is now a 2D probability distribution and the variables are correlated; this is clear as e.g. a high number for n_6 leaves less possibilities for n_5 given a fixed total of N throws. This generalises to a ‘multinomial’ distribution for more possible outcomes

$$M(n_i; p_i, N) = N! \prod_i \frac{p_i^{n_i}}{n_i!}$$

where i runs over the outcomes, p_i is the probability of each outcome (with the constraint that $\sum_i p_i = 1$) and n_i are the observed numbers for each outcome, with $N = \sum_i n_i$ being the total number of trials.

11 Poisson distribution

The Poisson distribution describes the probability of obtaining a number of occurrences n given an average number μ , where the probability of each occurrence is independent. The distribution is

$$P(n; \mu) = \frac{\mu^n e^{-\mu}}{n!}$$

Here, n is unbounded above and so can take any discrete value from 0 to ∞ . Some examples: the number of photons from a patch of sky in a particular photograph, the number of lightening strikes per hour during a storm, the number of buses passing through a road junction, or the number of car crashes on the M1 per year. It differs from the binomial because the occurrences happen in a continuum, i.e. there is no specific number of trials and we can’t say how often we didn’t get an occurrence; e.g. it is meaningless to ask how many times a bus *didn’t* go through the junction.

It is again easy to check the probabilities sum to one by using the standard expansion of an exponential

$$e^\mu = 1 + \mu + \frac{1}{2}\mu^2 + \dots = \sum_{n=0}^{\infty} \frac{\mu^n}{n!}$$

Hence multiplying by $e^{-\mu}$ gives

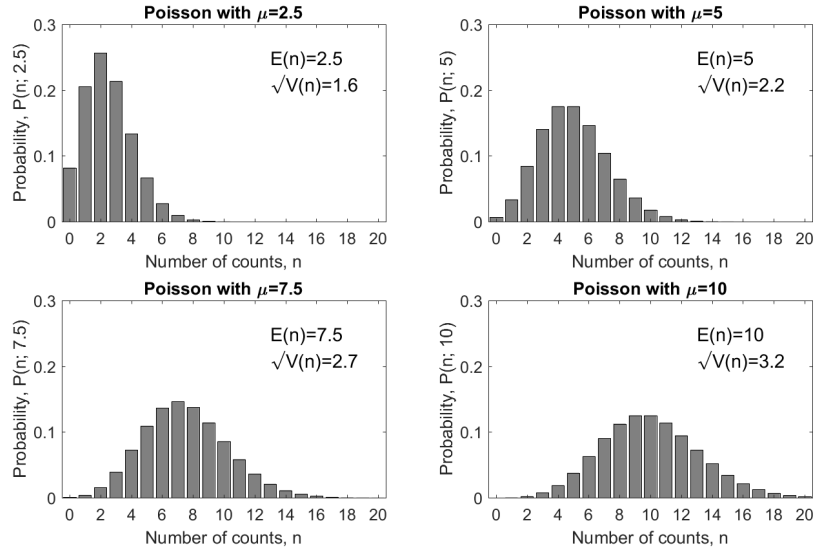
$$e^\mu e^{-\mu} = 1 = \sum_{n=0}^{\infty} \frac{\mu^n e^{-\mu}}{n!} = \sum_{n=0}^{\infty} P(n; \mu)$$

as required.

The expectation value and variance of a Poisson distribution can be found (see the appendix) to be

$$E(n) = \mu, \quad V(n) = \mu$$

which means the standard deviation is $\sqrt{\mu}$. The expression for the expectation value shows that μ is indeed the average of the number of occurrences seen, as originally stated. Note, the parameter μ , being the average number, is not necessarily an integer, while the outcome n is the number seen and must be an integer. Some examples of Poisson distributions with the expectation value and standard deviation marked on them are shown below. It can be seen how the expectation value increases and the distribution spreads out as μ increases, similar to what we saw for increasing N in the binomial distribution.



In many cases, the Poisson mean μ is basically calculated from the rate of some process being measured for a fixed time or area. Take the example of photons from a patch of sky; if the weather is stable, there should be a constant rate of photons per unit time, λ . Hence, the average number of photons in time t would be $\mu = \lambda t$. Therefore, knowing the rate allows us to find the probabilities for n occurrences for whatever length of time t that we run our experiment for, i.e.

$$P(n; \mu = \lambda t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}$$

Similarly, if we know the average density of buses per km^2 in London, ρ , then we can calculate the probability of n buses in an area A using $\mu = \rho A$. It is common to see the Poisson distribution written in forms like these.

12 Poisson as a limit of binomial

One critical difference between a Poisson and a binomial is that in the Poisson, the number of occurrences n can go to infinity, while in a binomial, it has a maximum of N . However, if p is small in the binomial, then the probability of getting n nearly as large as N is very small and then the upper limit becomes irrelevant. Let's consider the binomial as N gets very large and p gets very small, but with the constraint that pN remains finite. The binomial is

$$B(n; p, N) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n}$$

Consider the numerator of the binomial coefficient $N!$. This is

$$N! = N(N-1)\dots(N-n+1)(N-n)(N-n-1)\dots 1 = N(N-1)\dots(N-n+1) \times (N-n)!$$

Hence, the numerator includes the $(N-n)!$ in the denominator as a factor and so these can be cancelled. (In fact, this is often the best way to calculate the binomial coefficients.) This leaves the product of n terms above, so

$$B(n; p, N) = \frac{N(N-1)\dots(N-n+1)}{n!} p^n (1-p)^{N-n}$$

In the limit of large N and small p , then the only values of n with non-negligible probabilities are small, i.e. $n \ll N$. Hence, all terms like $N-1$ or $N-n$ can be approximated to N , so the binomial is approximately

$$B(n; p, N) \approx \frac{N^n}{n!} p^n (1-p)^N$$

It is a standard result that in the limit of $N \rightarrow \infty$ and $p \rightarrow 0$, such that pN remains finite, then

$$(1-p)^N \rightarrow e^{-pN}$$

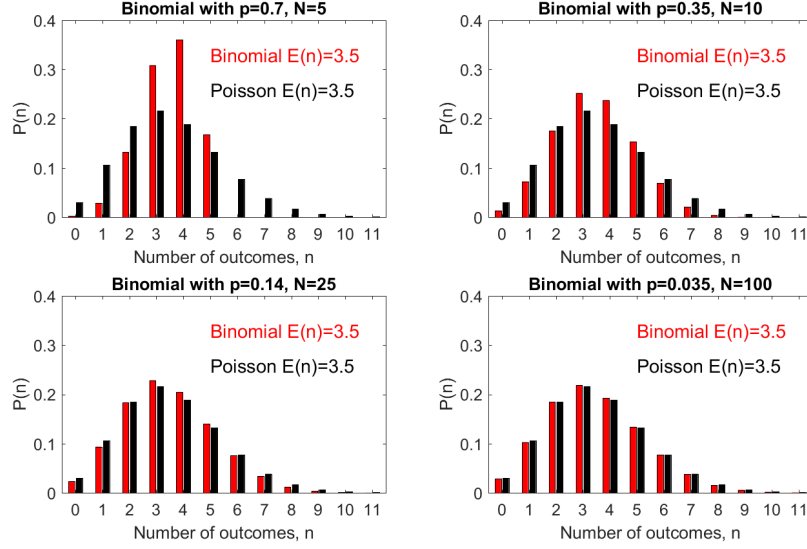
so

$$B(n; p, N) \approx \frac{(pN)^n}{n!} e^{-pN} = P(n; \mu = pN)$$

Hence, in this limit, the binomial becomes equal to the Poisson distribution, where the average value is $\mu = pN$, i.e. the same as the average of the binomial distribution. This is shown graphically in the figure below where the distributions become similar as N in the binomial gets larger. See also how you can have different binomial distributions with the same expectation value but different parameters, and different variances and standard deviations.

As stated above, the number of nuclear decays from a source will follow a binomial in principle, as we cannot see more decays than the number of nuclei we have. However, any macroscopic piece of radioactive material has a finite but huge number of nuclei, e.g. of order Avogadro's number. Hence, unless the decay is very rapid, the number of decays observed will be very small compared with the total number, so in this case, the number observed is a very good approximation to a Poisson distribution.

Conversely, in fact, it is very rare to ever get a 'real' Poisson distribution in principle. In the cases mentioned above, they are all in fact fundamentally binomial distributions but are extremely good approximations to Poisson distributions. For the number of photons from a patch of sky, the total possible number would be huge but cannot be infinite as there is only a finite amount of energy in the Universe. Lightning strikes must be limited by the amount of static generated on the clouds. Finally, the number of buses or car crashes must be limited to the total number of vehicles in the UK. However, in all cases, the number observed is minute compared with the maximum possible, so they are all very accurately described by a Poisson distribution.



13 Non-examinable: Appendix

13.1 Expectation value and variance of a binomial distribution

The expectation value of a binomial distribution is

$$\begin{aligned}
 E(n) = \langle n \rangle &= \sum_{n=0}^N n B(n; p, N) = \sum_{n=1}^N n \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} \\
 &= Np \sum_{n=1}^N \frac{(N-1)!}{(n-1)!(N-n)!} p^{n-1} (1-p)^{N-n} \\
 &= Np \sum_{n=1}^N \frac{(N-1)!}{(n-1)![(N-1)-(n-1)]!} p^{n-1} (1-p)^{(N-1)-(n-1)}
 \end{aligned}$$

where in the first line we have made use of the fact that the $n = 0$ term in the sum is 0, hence we can sum from $n = 1$ onwards. Using the substitution $m = n - 1$ and $M = N - 1$ we obtain

$$E(n) = \langle n \rangle = Np \sum_{m=0}^M \frac{M!}{m!(M-m)!} p^m (1-p)^{M-m} = Np \sum_{m=0}^M B(m; p, M) = Np$$

as the sum above equals 1, being the sum of the terms of a binomial distribution which is normalized to a total probability of 1.

To compute the variance, we start by noticing that

$$V(n) = \langle n^2 \rangle - \langle n \rangle^2 = \langle n(n-1) + n \rangle - \langle n \rangle^2 = \langle n(n-1) \rangle + \langle n \rangle - \langle n \rangle^2$$

Hence, as we know $E(n) = \langle n \rangle$ already, we only need to compute $\langle n(n-1) \rangle$. Using a similar method to previously

$$\begin{aligned}
 \langle n(n-1) \rangle &= \sum_{n=0}^N n(n-1) \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} = \sum_{n=2}^N \frac{N!}{(n-2)!(N-n)!} p^n (1-p)^{N-n} \\
 &= N(N-1)p^2 \sum_{n=2}^N \frac{(N-2)!}{(n-2)!(N-n)!} p^{n-2} (1-p)^{N-n}
 \end{aligned}$$

$$= N(N-1)p^2 \sum_{n=2}^N \frac{(N-2)!}{(n-2)![(N-2)-(n-2)]!} p^{n-2}(1-p)^{(N-2)-(n-2)}$$

Substituting $m = n - 2$ and $M = N - 2$ we obtain

$$\langle n(n-1) \rangle = N(N-1)p^2 \sum_{m=0}^M \frac{M!}{m!(M-m)!} p^m (1-p)^{M-m} = N(N-1)p^2 \sum_{m=0}^M B(m; p, M) = N(N-1)p^2$$

Thus

$$V(n) = \langle n(n-1) \rangle + \langle n \rangle - \langle n \rangle^2 = N(N-1)p^2 + Np - (Np)^2 = N^2p^2 - Np^2 + Np - N^2p^2 = Np(1-p)$$

and hence the standard deviation is $\sqrt{Np(1-p)}$.

13.2 Expectation value and variance of a Poisson distribution

Using the well-known properties of an exponential

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad \text{and} \quad \frac{d}{dx}(e^x) = e^x$$

this means

$$\frac{d}{dx}(e^x) = \sum_{n=0}^{\infty} n \frac{x^{n-1}}{n!} = e^x$$

Multiplying both sides by x gives

$$xe^x = \sum_{n=0}^{\infty} n \frac{x^n}{n!}$$

The expectation value of a Poisson distribution is therefore given by

$$E(n) = \langle n \rangle = \sum_{n=0}^{\infty} n P(n; \mu) = \sum_{n=0}^{\infty} n \frac{\mu^n}{n!} e^{-\mu} = e^{-\mu} \left(\sum_{n=0}^{\infty} n \frac{\mu^n}{n!} \right) = e^{-\mu} (\mu e^{\mu}) = \mu$$

To compute the variance, we need $V(n) = \langle n^2 \rangle - \langle n \rangle^2$. Using a similar trick

$$\frac{d^2}{dx^2}(e^x) = \sum_{n=0}^{\infty} n(n-1) \frac{x^{n-2}}{n!} = e^x$$

and hence

$$x^2 e^x = \sum_{n=0}^{\infty} n(n-1) \frac{x^n}{n!} = \sum_{n=0}^{\infty} (n^2 - n) \frac{x^n}{n!} = \sum_{n=0}^{\infty} n^2 \frac{x^n}{n!} - x e^x$$

Therefore

$$\sum_{n=0}^{\infty} n^2 \frac{x^n}{n!} = x^2 e^x + x e^x$$

We can now compute

$$\langle n^2 \rangle = \sum_{n=0}^{\infty} n^2 P(n; \mu) = \sum_{n=0}^{\infty} n^2 \frac{\mu^n}{n!} e^{-\mu} = e^{-\mu} \left(\sum_{n=0}^{\infty} n^2 \frac{\mu^n}{n!} \right) = e^{-\mu} (\mu^2 e^{\mu} + \mu e^{\mu}) = \mu^2 + \mu$$

so that

$$V(n) = \mu^2 + \mu - \mu^2 = \mu$$

Hence, the standard deviation is $\sqrt{\mu}$.

Statistics of Measurement

Lecture 3 - Continuous probability densities

Heather Graven, 16 May 2023

14 General properties

In the previous lecture, we saw probability distributions where the outcomes were discrete, labelled as x_i , and had particular probabilities associated with them. However, it is often the case that we do not measure discrete quantities (like the number of nuclear decays in a second) but instead measure continuous quantities (like a voltage). The outcome of such measurements can take any value in a continuum, i.e. the sample space is continuous. We will again call the random variable x , although it is not necessarily a position in space.

The probability of any particular exact value such as $x = 4.3534679265873\dots$ is clearly negligible. Hence, in the continuous case, the probabilities are given by a **probability density function** (PDF), here denoted by $\rho(x)$. This is defined such that the (infinitesimal) probability of finding the outcome value in the range x to $x + dx$ is $\rho(x) dx$ and so a specific value (like the one above) has an infinitesimally small probability. To get a finite probability, then the probability density function has to be integrated over some range. Specifically, the probability of getting x within the range $x_1 \leq x \leq x_2$ is

$$P(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} \rho(x) dx$$

Note, since the probability is dimensionless, $\rho(x)$ has dimensions $[x^{-1}]$.

The integral of probability density over the whole possible range of x must be 1, akin to the normalisation requirement on the discrete sum of probabilities:

$$\sum_i P_i = 1 \rightarrow \int_{-\infty}^{\infty} \rho(x) dx = 1$$

The integral may have narrower limits if the range of x is restricted. An important physics example of this is in Quantum Mechanics, where the modulus squared of the wavefunction gives the probability density for finding the particle at position x , i.e. $\rho(x) = |\psi(x)|^2$ and the normalisation condition can be written as

$$\int_{-\infty}^{\infty} |\psi(x)|^2 dx = 1$$

which ensures that the particle is bound to be found somewhere. We will discuss several other probability density functions which commonly arise in physics below.

We also need to be able to calculate the expectation value and variance of a continuous probability density function. In the case of a continuous PDF, the probability is given by the continuous $\rho(x) dx$ rather than the discrete P_i , so the expression for the expectation value goes to

$$E(x) = \sum_i x_i P_i \rightarrow \int_{-\infty}^{\infty} x \rho(x) dx$$

Similarly, if we remember the expression for the variance is the expectation value for the squared difference from the mean

$$V(x) = E((x - \langle x \rangle)^2) = \langle x^2 \rangle - \langle x \rangle^2$$

then this goes from discrete to continuous as

$$V(x) = \sum_i x_i^2 P_i - \left(\sum_i x_i P_i \right)^2 \rightarrow \int_{-\infty}^{\infty} x^2 \rho(x) dx - \left(\int_{-\infty}^{\infty} x \rho(x) dx \right)^2$$

The standard deviation is again given by the square root of the variance.

15 Changing variables

For discrete distributions, then for (at least monotonic) functions $y = y(x)$, the probability of getting $y_i = y(x_i)$ is the same as getting x_i . However, when we deal with probability densities, then we have an integral for the probability and so the density function changes if we change the variable. If the probability density in x is $\rho_x(x)$, then as stated above, the probability of x being between x_1 and x_2 is given by

$$P(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} \rho_x(x) dx$$

If we want to change variables, then we need to express this as an integral over y . Using $dx = (dx/dy) dy$, and writing the limits as $y_1 = y(x_1)$, $y_2 = y(x_2)$, then this is

$$P(y_1 \leq y \leq y_2) = \int_{y_1}^{y_2} \frac{dx}{dy} \rho_x(x) dy$$

and hence, since changing variables cannot change the probability, we have

$$\rho_y(y) = \left| \frac{dx}{dy} \right| \rho_x(x) = \left| \frac{dy}{dx} \right|^{-1} \rho_x(x)$$

where the right-hand side terms should be expressed in terms of y . Note the modulus of the derivative, as a probability density has to be positive. Effectively, we are changing from dx to dy and hence for a PDF of more than one variable, this generalises to multiplying by the Jacobian.

For example, if $\rho_x(x) = x^3$ and $y = x^2$, then $dy/dx = 2x$ and so

$$\rho_y(y) = \frac{1}{2x} x^3 = \frac{x^2}{2} = \frac{y}{2}$$

Note it is *not* correct to state that since $x = y^{1/2}$ and $\rho_x(x) = x^3$ then $\rho_y(y) = (y^{1/2})^3 = y^{3/2}$. Therefore, changing variables is not quite as simple as plugging the expression for new variable into the original probability density.

Finally, one common source of error is non-monotonic functions, where different values of x give the same value of y . These have to be handled by dividing up the PDF in x into monotonic pieces, applying the above, and then adding the resulting PDFs to get the total PDF for y .

16 Continuous uniform density function

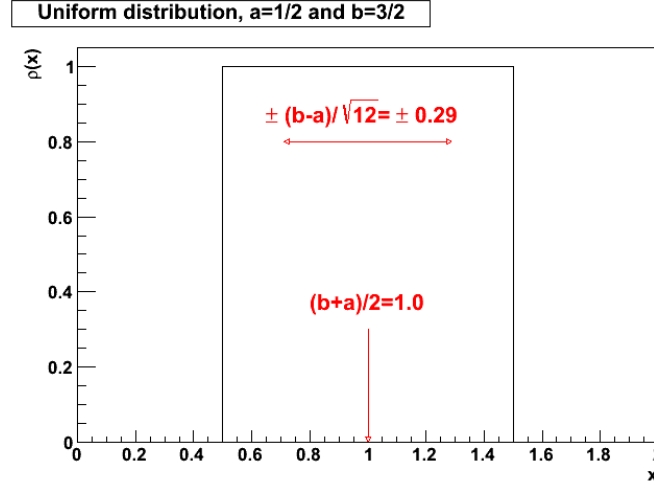
This is the continuous equivalent of the discrete uniform probability distribution which we met in Lecture 1. The continuous uniform probability density function has the same value ρ_0 anywhere within the allowed range of x . Note, the x range cannot be $\pm\infty$ as it would not be possible to normalise the PDF. If the range of x is $a \leq x \leq b$ then we have

$$\begin{aligned} \rho(x; a, b) &= \rho_0 & \text{for } a \leq x \leq b \\ &= 0 & \text{otherwise} \end{aligned}$$

The normalisation requirement means

$$\int_{-\infty}^{\infty} \rho(x) dx = \int_a^b \rho(x) dx = \int_a^b \rho_0 dx = \rho_0 [x]_a^b = \rho_0(b-a) = 1$$

so $\rho_0 = 1/(b-a)$. The expectation value is intuitively the middle of the range, $(b+a)/2$ (for detail see the appendix) and the variance can be calculated to be $V(x) = (b-a)^2/12$ (see the appendix) so the standard deviation is $(b-a)/\sqrt{12}$. An example is shown below.



This gives a useful way to estimate an error when making a digitised measurement. For example, a digital voltmeter gives a reading of $V = 4.1$ V. In the absence of other errors, what this means is the the actual voltage is somewhere in the range $V = 4.05$ to 4.15 V. Hence, if we treat it as a uniform distribution within that range, then the standard deviation would be

$$\text{Std. Dev.} = \frac{4.15 - 4.05}{\sqrt{12}} = \frac{0.1}{3.46} \sim 0.03 \text{ V}$$

which could be used as the estimated measurement error.

17 Exponential density function

The exponential PDF appears in many places in physics. It arises whenever there is a Poisson distribution based on a constant rate, as discussed in the previous lecture. In the case of a constant rate in time, the probability density for the time between each occurrence is given by the exponential distribution.

To see why, say the average rate of occurrences is λ , so the average number in time t is λt . We know the probability of no occurrences between time 0 and t from the Poisson distribution to be $P(0; \lambda t) = e^{-\lambda t}$. The probability of one occurrence between time t and $t + dt$ is $P(1; \lambda dt) = (\lambda dt)e^{-\lambda dt}$. However, for infinitesimal dt , the exponential goes to $e^0 = 1$, so the probability is λdt , as you might expect. Hence, the probability of the next occurrence being between t and $t + dt$ is the probability of no occurrences up to t , times the probability of an occurrence in t to $t + dt$, which is $\lambda dt e^{-\lambda t}$. Therefore, the PDF is

$$\rho(t; \lambda) = \lambda e^{-\lambda t}$$

This was for the example of a rate in time; it could also be e.g. the distance between places where crashes happened on the M1, so generically, we will write

$$\rho(x; \lambda) = \lambda e^{-\lambda x}$$

where λ is the average number per unit of x . The exponential distribution is valid for $0 \leq x \leq \infty$. It is also commonly expressed with a different parametrisation

$$\rho(x; a) = \frac{1}{a} e^{-x/a}$$

where a is the average separation in units of x , e.g. the average time between occurrences. Clearly $\lambda = 1/a$ and the average number expected in a period Δx is $\mu = \lambda \Delta x = \Delta x/a$. The factor in front of the exponential ensures the distribution is correctly normalised, so

$$\int_0^\infty \lambda e^{-\lambda x} dx = [-e^{-\lambda x}]_0^\infty = 1 \quad \text{or} \quad \int_0^\infty \frac{1}{a} e^{-x/a} dx = [-e^{-x/a}]_0^\infty = 1$$

For the exponential distribution, the expectation value is

$$E(x) = \int_0^\infty x \lambda e^{-\lambda x} dx$$

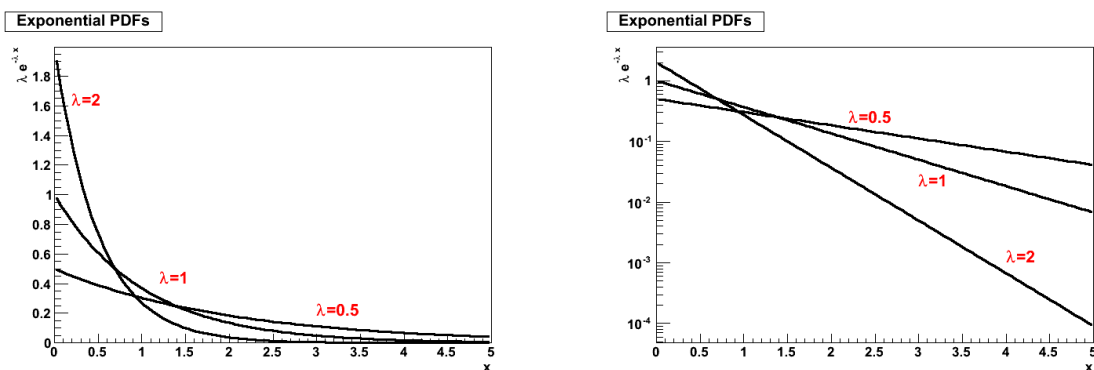
As shown in the appendix, integrating by parts gives

$$E(x) = \frac{1}{\lambda} = a$$

as expected if a is the average separation. The variance is also calculated by integrating by parts to give

$$V(x) = \frac{1}{\lambda^2} = a^2 \quad \text{and} \quad \sigma = \frac{1}{\lambda} = a = E(x)$$

Some examples of exponential PDFs with both a linear and logarithmic y axis are shown below. It can be seen how larger values of λ indicate faster rates and shorter intervals between occurrences.



Example: you are waiting at a bus stop. There are 2.5 buses per hour on average but the traffic variation means they turn up at random times. The average rate is $\lambda = 2.5/\text{hour}$ (i.e. the average time between each is $a = 24$ minutes). What is the probability you would wait for at least 30 minutes? This is given by

$$P(0.5 \leq t \leq \infty) = \int_{0.5}^\infty \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_{0.5}^\infty = e^{-0.5\lambda} = e^{-1.25} = 0.287$$

18 Gaussian density function

The Gaussian PDF (also called the *normal* or *bell shape* PDF) is perhaps the most important continuous distribution. It is used in many situations involving continuous random variables.

The reason for this, and the justification for the functional shape, will be given in the next lecture when we study the **Central Limit Theorem**. The Gaussian PDF is given by

$$G(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

where the range of x is $-\infty < x < \infty$ and the parameters μ and σ directly control the mean and standard deviation, respectively. For a Gaussian, the probability is proportional to the exponential of the square of the difference from the mean. Therefore, the probability is highest at the mean value and then decreases moving away from the mean in either direction, where this decrease follows the exponential of the square of the difference from the mean. The variance appears in the exponent and governs the steepness of the decrease in probability moving away from the mean, as shown in the example plots below. The factors in front normalise the distribution such that the area under the curve is equal to one, as required for a probability distribution

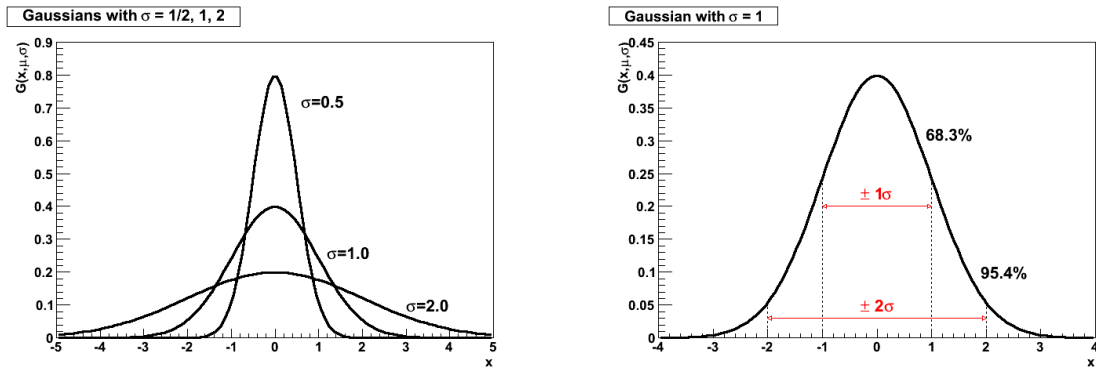
$$\int_{-\infty}^{\infty} G(x; \mu, \sigma) dx = 1$$

It is common to use the **standard score** $Z = (x - \mu)/\sigma$ such that $dx = \sigma dZ$ and hence

$$G(Z) = \frac{1}{\sqrt{2\pi}} e^{-Z^2/2}$$

Converting to Z makes the expression general, and its value indicates how far x is from the mean, relative to the standard deviation, e.g. for x within one standard deviation of the mean Z will be between -1 and 1. As stated above (and calculated in the appendix)

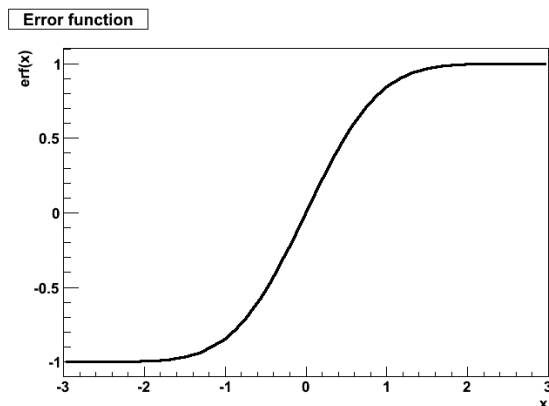
$$E(x) = \mu, \quad V(x) = \sigma^2 \quad \text{and} \quad \text{Standard deviation} = \sigma$$



It is often the case that we are trying to measure something, e.g. the length of a rod, and we get slightly different values each time we do it. These most commonly have a Gaussian distribution with a mean at the true value of the length (if our ruler is unbiased), but each individual measurement is spread around the mean with the standard deviation σ . The measurements are then said to have an uncertainty (or error) and the size of the uncertainty reflects the spread of the values. By convention, the uncertainty is specified by the standard deviation σ . Note, however, that this does not absolutely contain all possible values of x ; indeed, the tails of the Gaussian go out to infinity so no finite range can contain all values. To find the probability of x being within $\pm 1\sigma$ of μ , we would need to calculate $\int_{\mu-\sigma}^{\mu+\sigma} G(x; \mu, \sigma) dx$. Unfortunately, the integral over a finite range of a Gaussian cannot be done analytically. However, the closely related ‘error function’ (written as ‘erf’) is defined to be

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-y^2} dy$$

and this is normally implemented in most software packages, e.g. in `scipy.special` in python. With a change of variables, it is possible to express Gaussian integrals in terms of $\text{erf}(x)$ and evaluate them numerically. A plot of the erf function is shown below.



As an example of using the error function, again consider calculating the fraction of the Gaussian contained within the $\pm 1\sigma$ range. This is

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{\mu-\sigma}^{\mu+\sigma} e^{-(x-\mu)^2/2\sigma^2} dx = \frac{1}{\sqrt{2\pi}} \int_{-1}^1 e^{-Z^2/2} dZ$$

Changing variables from x to $y = (x - \mu)/\sqrt{2}\sigma = Z/\sqrt{2}$ means $dZ = \sqrt{2} dy$ and the limits are $y = \pm 1/\sqrt{2}$, so the integral becomes

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-y^2} \sqrt{2}\sigma dy = \frac{1}{\sqrt{\pi}} \int_{-1/\sqrt{2}}^{1/\sqrt{2}} e^{-y^2} dy = \frac{2}{\sqrt{\pi}} \int_0^{1/\sqrt{2}} e^{-y^2} dy$$

where the last equality follows as the function being integrated is even. Hence this gives

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{\mu-\sigma}^{\mu+\sigma} e^{-(x-\mu)^2/2\sigma^2} dx = \text{erf}(1/\sqrt{2}) = \text{erf}(0.707) = 0.683$$

This means that the probability of x being within this range of $\pm 1\sigma$ is 68.3%. This is true of all Gaussian distributions and one of the few numbers you are expected to remember from this course. In addition to numerical calculations using the error function, tables of numerical values relating to the Gaussian distribution are widely available.

Now that we know the probability of x being within the range of $\pm 1\sigma$ is 68.3%, we can reflect on the convention of using the standard deviation for uncertainty. When we quote the uncertainty as $\pm 1\sigma$ we are quoting the range for which the true value of x is within with probability 68.3%, so a pretty good chance but still a substantial chance it could be outside this range. This convention is somewhat arbitrary and if we wanted to be more conservative, then we could specify the uncertainty as $\pm 2\sigma$, which would give a probability of 95.4%, or $\pm 3\sigma$ for 99.7%. Alternatively, we could set the probability we want and then calculate the range as a multiple of the standard deviation. Some common choices are given in the table below. These are all examples of ‘confidence intervals’ which we will discuss further in lecture 8; for example, we would say we are 68.3% confident that x will be in the $\pm 1\sigma$ interval. Note that quoting $\pm 1\sigma$ as the uncertainty is by far the most common, but it is good practice to always say what the uncertainty refers to. This is particularly important when using something other than $\pm 1\sigma$.

$\pm\sigma$	Fraction	Fraction	$\pm\sigma$
1	0.683	0.90	1.64
2	0.954	0.95	1.96
3	0.997	0.99	2.58

19 Cumulative distribution function

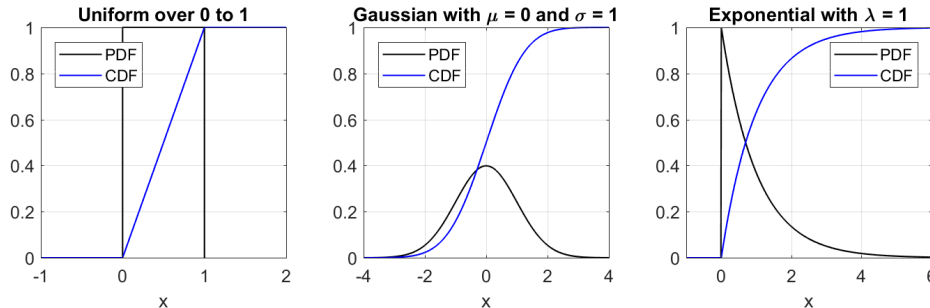
The cumulative distribution function (CDF) is another useful way of representing a probability distribution. The cumulative distribution function $F(x)$ is the probability that the random variable X takes on a value that is less than or equal to x

$$F(x) = P(X \leq x)$$

If X is a continuous random variable then the CDF is given by the integral of the PDF $\rho(x)$ from $-\infty$ (or the lower limit of the distribution) to x

$$F(x) = \int_{-\infty}^x \rho(t) dt$$

It can be seen that $F(x)$ is an anti-derivative of $\rho(x)$. As x tends to $-\infty$ (or the lower limit) then $F(x)$ tends to zero and as x tends to ∞ (or the upper limit) then $F(x)$ tends to one, i.e. the entire pdf is integrated. The plots below give examples of CDFs for the uniform, exponential and Gaussian distributions. Note that the CDF for the Gaussian (given by $\frac{1}{2}(1 + \operatorname{erf}(\frac{x-\mu}{\sigma\sqrt{2}}))$) resembles the error function above, as expected. CDFs can also be created for discrete distributions by summing the probabilities for all possible values less than x .



20 Non-examinable: Appendix

20.1 Convolution

With discrete distributions the outcome is unambiguous; one of the discrete x_i is observed. However, with continuous distributions, then the variables can have a measurement uncertainty in addition to the ‘underlying’ distribution.

For example, the exponential distribution is used to measure the rate of nuclear decays. This involves measuring the time t until the next decay. If the measurement of t itself has a negligible uncertainty, then we will see a pure exponential. However, if the measured value of t has a Gaussian uncertainty with a significant width compared with the lifetime, then the measurements will have a significant smearing and the exponential will be distorted. Indeed, for a very short lifetime, the uncertainty may mean the apparent measured time is negative!

Consider a small part of the exponential, between t' and $t' + dt'$. If there was no uncertainty on the time measurement, then we would observe t' directly. However, t' is smeared by the

Gaussian to give us a measured time t . The probability of the pure exponential being between t' and $t' + dt'$ is $\lambda e^{-\lambda t'} dt'$. This will be smeared by a Gaussian, which will of course be centred on t' . Hence, the probability density for observed time t around the exponential region at t' is

$$G(t; t', \sigma) \lambda e^{-\lambda t'} dt'$$

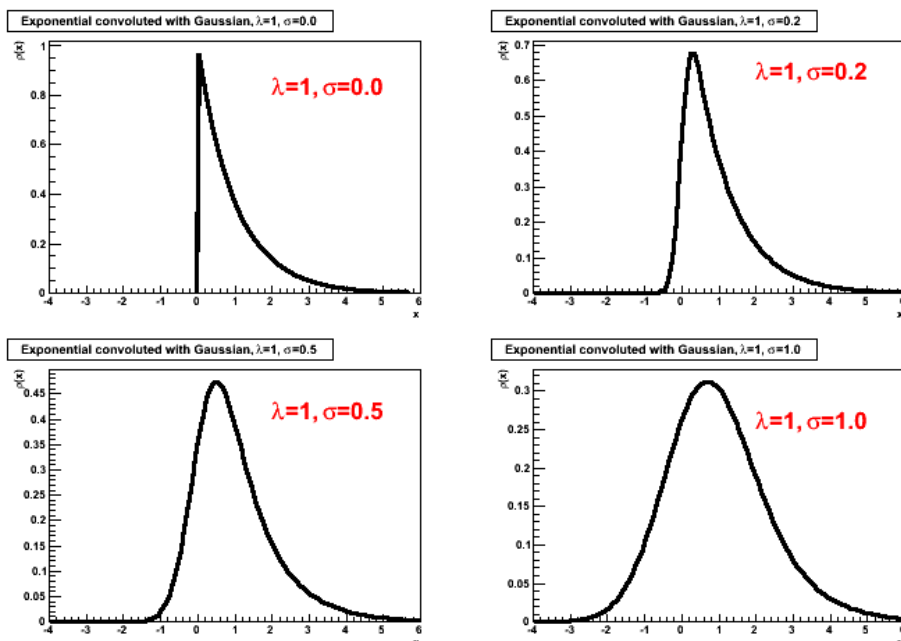
We only see t , so we need to integrate over all possible t' to get the PDF in terms of t only, i.e.

$$\rho(t; \lambda, \sigma) = \int_0^\infty G(t; t', \sigma) \lambda e^{-\lambda t'} dt'$$

This is messy but doable and gives

$$\rho(t; \lambda, \sigma) = \frac{\lambda e^{\lambda^2 \sigma^2 / 2}}{2} e^{-\lambda t} \left[1 - \operatorname{erf} \left(\frac{\lambda \sigma^2 - t}{\sigma \sqrt{2}} \right) \right]$$

This process is called 'convolution'. Convolutions of an exponential distribution with a Gaussian distribution are shown below for various values of σ .



We can also do a convolution of a Gaussian with another Gaussian, as would be seen if there are two sources of uncertainty in a measurement, e.g. when measuring a circuit voltage and this depending on a fluctuating temperature as well as on the uncertainty from the voltmeter. In general the two Gaussians will have different widths σ_1 and σ_2 . This convolution is also messy but the answer is very simple; it is just another Gaussian. The mean is unchanged (as you would expect, given that you are convoluting two symmetric distributions) and the overall width is

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2}$$

Clearly, the total width is larger than either contribution. This should look familiar from the propagation of errors formula for $z = x + y$; it is effectively the same thing because for every value of x (which has a Gaussian distribution), the sum (i.e. z) is also smeared by the Gaussian in y .

20.2 Expectation value and variance of a uniform distribution

The expectation value of a continuous uniform distribution is

$$E(x) = \int_a^b \frac{x}{b-a} dx = \frac{1}{2(b-a)} \left[x^2 \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{b+a}{2}$$

i.e. the centre of the distribution, as would be expected.

For the variance, we need the average of x^2 . This is

$$\langle x^2 \rangle = \int_a^b \frac{x^2}{b-a} dx = \frac{1}{3(b-a)} \left[x^3 \right]_a^b = \frac{b^3 - a^3}{3(b-a)}$$

Hence the variance is

$$\begin{aligned} V(x) &= \langle x^2 \rangle - \langle x \rangle^2 = \frac{b^3 - a^3}{3(b-a)} - \frac{(b+a)^2}{4} = \frac{4(b^3 - a^3) - 3(b+a)^2(b-a)}{12(b-a)} \\ &= \frac{4b^3 - 4a^3 - 3b^3 - 6b^2a - 3ba^2 + 3b^2a + 6ba^2 + 3a^3}{12(b-a)} \\ &= \frac{b^3 - 3b^2a + 3ba^2 - a^3}{12(b-a)} = \frac{(b-a)^3}{12(b-a)} = \frac{(b-a)^2}{12} \end{aligned}$$

and hence the standard deviation is $(b-a)/\sqrt{12}$.

20.3 Expectation value and variance of an exponential distribution

Integrating by parts, the expectation value is

$$E(x) = \int_0^\infty \lambda x e^{-\lambda x} dx = \int_0^\infty e^{-\lambda x} dx - \left[x e^{-\lambda x} \right]_0^\infty = -\frac{1}{\lambda} \left[e^{-\lambda x} \right]_0^\infty = \frac{1}{\lambda}$$

For the variance, we need the average of x^2 . Again by parts, this is

$$\langle x^2 \rangle = \int_0^\infty \lambda x^2 e^{-\lambda x} dx = 2 \int_0^\infty x e^{-\lambda x} dx - \left[x^2 e^{-\lambda x} \right]_0^\infty = \frac{2}{\lambda^2}$$

and so

$$V(x) = \langle x^2 \rangle - \langle x \rangle^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

and hence the standard deviation is $1/\lambda$.

20.4 Expectation value and variance of a Gaussian distribution

Consider the expectation value of the standard score Z

$$E(Z) = \int_{-\infty}^{\infty} Z \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ = \frac{1}{\sqrt{2\pi}} \left[-e^{-Z^2/2} \right]_{-\infty}^{\infty} = 0$$

The expectation value of x is

$$\begin{aligned} E(x) &= \int_{-\infty}^{\infty} x \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} dx = \int_{-\infty}^{\infty} (\sigma Z + \mu) \frac{1}{\sigma\sqrt{2\pi}} e^{-Z^2/2} \sigma dZ \\ &= \sigma \int_{-\infty}^{\infty} Z \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ + \mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ \\ &= \mu \end{aligned}$$

since the first integral is $E(Z) = 0$ and the second is the PDF in terms of Z and so integrates to one.

Since $E(Z) = \langle Z \rangle = 0$, the variance of Z is simply

$$V(Z) = \int_{-\infty}^{\infty} Z^2 \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ - E(Z)^2 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} Z^2 e^{-Z^2/2} dZ = \langle Z^2 \rangle$$

Using integration by parts, this is

$$V(Z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-Z^2/2} dZ - \frac{1}{\sqrt{2\pi}} \left[Z e^{-Z^2/2} \right]_{-\infty}^{\infty} = 1$$

again using the normalisation of the PDF. The variance of x is then

$$\begin{aligned} V(x) &= \int_{-\infty}^{\infty} x^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2} dx - E(x)^2 = \int_{-\infty}^{\infty} (\sigma Z + \mu)^2 \frac{1}{\sigma\sqrt{2\pi}} e^{-Z^2/2} \sigma dZ - \mu^2 \\ &= \sigma^2 \int_{-\infty}^{\infty} Z^2 \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ + 2\sigma\mu \int_{-\infty}^{\infty} Z \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ + \mu^2 \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ - \mu^2 \\ &= \sigma^2 \end{aligned}$$

since the first integral is $V(Z) = 1$, the second is $E(Z) = 0$ and the third is the normalisation $= 1$.

Statistics of Measurement

Lecture 4 - The Central Limit Theorem

Heather Graven, 18 May 2023

21 Introduction

We have seen the expectation values and variances of the various discrete and continuous distributions we have met so far; these are summarised in the table below, using the notation used in the previous lectures.

		Expectation value	Variance	Standard deviation
Discrete	Binomial	pN	$Np(1-p)$	$\sqrt{Np(1-p)}$
	Poisson	μ	μ	$\sqrt{\mu}$
Continuous	Uniform	$(b+a)/2$	$(b-a)^2/12$	$(b-a)/\sqrt{12}$
	Exponential	$1/\lambda = a$	$1/\lambda^2 = a^2$	$1/\lambda = a$
	Gaussian	μ	σ^2	σ

In practice, it is typical that a variable will have a probability distribution that does not follow an ideal theoretical distribution because there may be multiple influences on that variable that have different distributions or multiple sources of error in the measurement. In addition, it is often the case that an experiment will take several measurements and compute the average. The average is clearly found from the sum of the measurements. The expected probability distribution of summed quantities can look very different from the underlying distribution from which the individual measurements are drawn. In this lecture, we will look at probability distributions for the sums of random variables.

22 Sums of random variables

Consider that we are going to sample N independent random variables x_i where the x_i are each from a separate PDF $\rho_i(x)$ which has an expectation value E_i and a variance V_i . Note, these individual distributions do not have to be Gaussian or even the same type of distribution. We will calculate a sum of them

$$X = \sum_i x_i$$

X is an example of a variable which is a function of only random variables, the x_i . Clearly, if we sampled another set of x_i , then since the x_i are random, we would have a different value of X . Hence, X , is itself a random variable.

We would like to know something about the probability distribution of X . For simplicity, we will initially look at summing just two random variables x_1 and x_2 with a PDF of $\rho(x_1, x_2)$. Consider the expectation values of the sum

$$\begin{aligned}
 E(X) &= \langle x_1 + x_2 \rangle = \int \int (x_1 + x_2) \rho(x_1, x_2) dx_1 dx_2 \\
 &= \int \int x_1 \rho(x_1, x_2) dx_1 dx_2 + \int \int x_2 \rho(x_1, x_2) dx_1 dx_2 \\
 &= \langle x_1 \rangle + \langle x_2 \rangle = E_1 + E_2
 \end{aligned}$$

i.e. the expectation value of the sum is simply the sum of the individual expectation values. Similarly, for the variance

$$\begin{aligned} V(X) &= \langle (X - \langle X \rangle)^2 \rangle = \langle [(x_1 + x_2) - (E_1 + E_2)]^2 \rangle = \langle [(x_1 - E_1) + (x_2 - E_2)]^2 \rangle \\ &= \langle (x_1 - E_1)^2 \rangle + \langle (x_2 - E_2)^2 \rangle + 2 \langle (x_1 - E_1)(x_2 - E_2) \rangle \end{aligned}$$

The variance of the sum is the sum of the variances (first two terms), plus the covariance (third term). If we have two independent (i.e. uncorrelated) variables, then the joint probability density is the product of the two separate probability densities

$$\rho(x_1, x_2) dx_1 dx_2 = \rho_1(x_1) \rho_2(x_2) dx_1 dx_2$$

In this case, the covariance term becomes

$$\begin{aligned} \langle (x_1 - E_1)(x_2 - E_2) \rangle &= \int \int (x_1 - E_1)(x_2 - E_2) \rho_1(x_1) \rho_2(x_2) dx_1 dx_2 \\ &= \int (x_1 - E_1) \rho_1(x_1) dx_1 \int (x_2 - E_2) \rho_2(x_2) dx_2 \end{aligned}$$

Each of these integrals is

$$\int (x_i - E_i) \rho_i(x_i) dx_i = \int x_i \rho_i(x_i) dx_i - E_i \int \rho_i(x_i) dx_i = E_i - E_i = 0$$

since E_i is the average of x_i . Conceptually, you can think of the covariance term being zero for independent random variables because a positive $(x_{1,i} - E_1)$ has an equal chance of being multiplied by a positive or negative $(x_{2,j} - E_2)$ and the average will be zero. Hence, following the derivation above and the conceptual reasoning, for independent variables

$$V(X) = \langle (x_1 - E_1)^2 \rangle + \langle (x_2 - E_2)^2 \rangle = V_1 + V_2$$

i.e. the variance of the sum is simply the sum of the variances. Note, the covariance term is only zero for two independent variables.

These results generalise to summing more than two independent random variables, such that

$$E(X) = \sum_i E_i, \quad V(X) = \sum_i V_i$$

Hence, in general, **both the expectation value and the variance of the sum is just the sum of the individual expectation values and variances** for independent random variables.

It turns out there are other quantities besides the expectation value and variance which can be used to characterise a distribution. The expectation value is linear in the x_i , while the variance is quadratic. There are further similar quantities for higher orders; the third order (or cubic) measure is called **skewness** S , which relates to the asymmetry of the distribution, while the fourth order (or quartic) is **(extended) kurtosis** K , which compares the density in the peak of the distribution vs the tails of the distribution relative to the Gaussian distribution, and so on. The exact definitions are given in the appendix. These higher order quantities are also all the sums of the corresponding quantities for the individual distributions for independent random variables, i.e. $S(X) = \sum_i S_i$, $K(X) = \sum_i K_i$, etc.

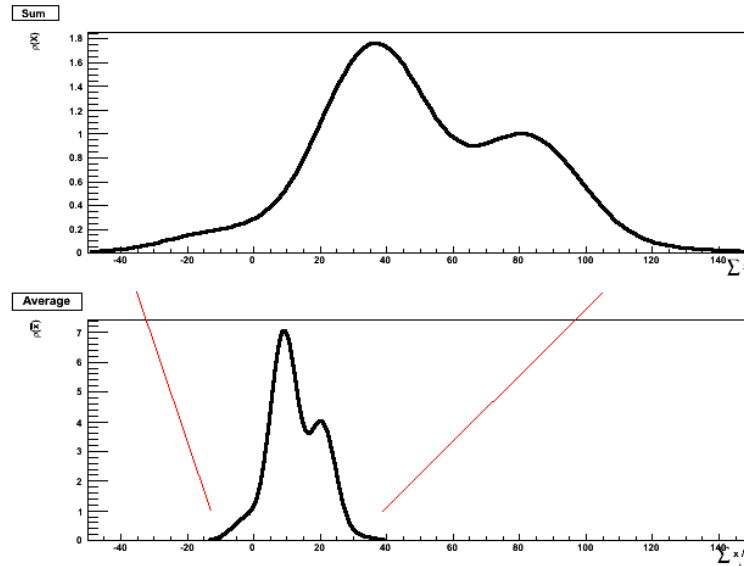
23 The Central Limit Theorem

The Central Limit Theorem (CLT) is a very general theorem which applies to any such cases. It basically says that a sum (and hence also the average) of N independent random variables will tend to have a Gaussian PDF as N gets large, irrespective of the original distributions of the measurements. This may seem bizarre, but you will see it is true.

We are interested in the limit of adding a very large number N of independent random variables x_i . For the sum X , all the quantities mentioned (expectation value, variance, skewness, etc) will generally keep getting bigger and bigger as N increases, as there are more and more contributions to the sums. Hence, these quantities all go to infinity as N gets large, which can obscure what the result is. It is mathematically less tricky to see what is happening if we consider the average of the x_i rather than the sum. The average is

$$\bar{x} = \frac{1}{N} \sum_i x_i = \frac{X}{N}$$

It is therefore just the sum divided by N . This means whatever the probability density distribution is for X , then $\bar{x}$ has the same PDF, but with the x axis scaled down by N , as shown in the figure.



This means the expectation value of $\bar{x}$ is the expectation value of X divided by N , as shown explicitly by changing variables

$$E(\bar{x}) = \int \bar{x} \rho(\bar{x}) d\bar{x} = \int \frac{X}{N} \rho(X) dX = \frac{E(X)}{N}$$

It also means the standard deviation of $\bar{x}$ is that of X divided by N . However, this results in the variance of $\bar{x}$ being that of X divided by N^2 . In fact, this should be clear as the variance is quadratic in X . Similarly, the skewness is divided by N^3 , etc. Hence, when adding more and more values of x_i , then the N dependences of the various quantities go as

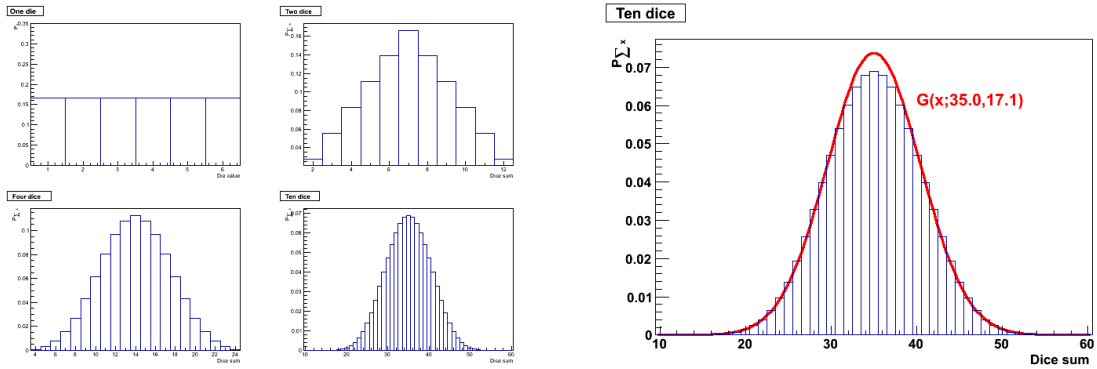
$$\begin{aligned} E(\bar{x}) &= E\left(\frac{X}{N}\right) = \frac{E(X)}{N} = \frac{\sum_i E_i}{N} && \text{goes as } \sim \frac{N}{N} \sim 1 \\ V(\bar{x}) &= V\left(\frac{X}{N}\right) = \frac{V(X)}{N^2} = \frac{\sum_i V_i}{N^2} && \text{goes as } \sim \frac{N}{N^2} \sim \frac{1}{N} \end{aligned}$$

$$\begin{aligned}
S(\bar{x}) &= S\left(\frac{X}{N}\right) = \frac{S(X)}{N^3} = \frac{\sum_i S_i}{N^3} && \text{goes as } \sim \frac{N}{N^3} \sim \frac{1}{N^2} \\
K(\bar{x}) &= K\left(\frac{X}{N}\right) = \frac{K(X)}{N^4} = \frac{\sum_i K_i}{N^4} && \text{goes as } \sim \frac{N}{N^4} \sim \frac{1}{N^3}
\end{aligned}$$

and so on. Therefore, in the large N limit, the leading term is the expectation value, and all other terms go to zero. This means as $N \rightarrow \infty$, then the only value for the $\bar{x}$ we can get will be the expectation value, as the width (and higher terms) are all zero. This is expected; we should get a measurement average equal to the expectation value in the limit of an infinite number of measurements. This means that in this limit, all measurement averages go to a single universal PDF shape, which is basically an infinitely thin spike (called a ‘Dirac delta function’) at the distribution average.

If we step back from the extreme limit for a moment, the next order term is the variance (since all other terms have a higher $1/N$ dependence). Hence, to this level of approximation, the distribution is that of a mean and a (small) variance, with all higher order terms zero. Again in this limit, all measurement averages have a universal PDF shape with only the expectation value and variance non-zero. The question is then; what distribution has the property that the skewness, kurtosis, etc, are all zero, as this will be the resulting distribution in all cases? The function which has this property turns out to be the Gaussian; this can be considered to be the defining property of the Gaussian distribution. Hence, no matter what the original distributions were, in the large N limit where the higher order terms above the variance can be ignored, then everything will look Gaussian.

As stated above, the sum X distribution and average $\bar{x}$ distribution are simply related by an N scaling. Hence, if the average is Gaussian, then so is the sum (albeit with a mean and width which go very large in the large N limit). As an example, the plot shows the sum of multiple dice throws (one, two, four and ten), and a comparison of the sum of ten dice throws with a Gaussian of the appropriate mean and variance.



Finally, note that if we are adding Gaussian-distributed random variables, then all the S_i , K_i , etc, higher order quantities are zero by definition. This means $S(X)$, $K(X)$, etc, are also all exactly zero, even for small N . Hence, the resulting distribution for X is by definition *exactly* Gaussian and the overall width will satisfy

$$\sigma^2 = \sum_i \sigma_i^2$$

This is another occurrence of **adding in quadrature**.

24 Sums of exponentially distributed random variables

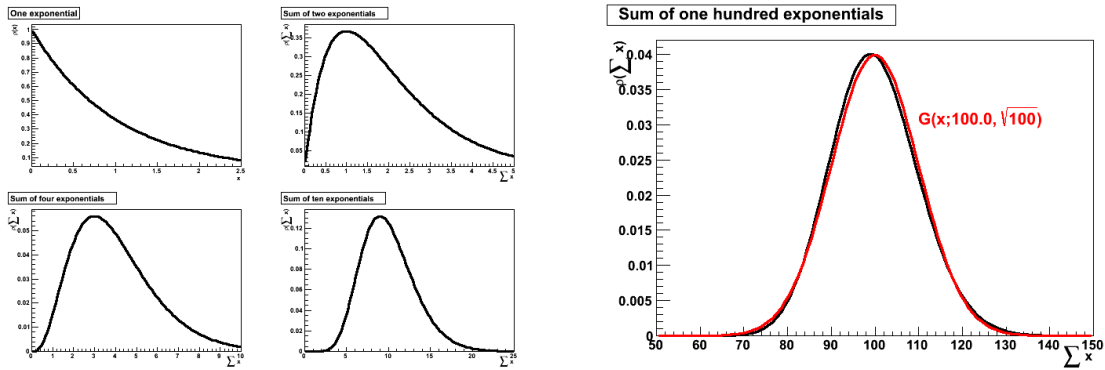
As another example of the Central Limit Theorem, consider summing two random variables drawn from exponential distributions. The sum can be anything from 0 to ∞ but there are many different combinations of the two values which will give the same sum. Since we are only interested in the sum, we have to integrate over all these combinations. The chance of getting a low value of the sum is relatively small as it requires both values to be small. Very large values in the sum are also unlikely, particularly as the exponential distribution falls off for large values. The resulting PDF for the sum of two random variables with the same exponential PDF is (see the appendix)

$$\rho(X) dX = \lambda^2 X e^{-\lambda X} dX$$

This generalises for the sum of N exponential random variables to give the PDF

$$\rho(X) dX = \lambda^N X^{N-1} e^{-\lambda X} dX$$

As N increases, this gives a PDF for the sum which looks more and more Gaussian; this is shown below. Again, the red curve gives the Gaussian with the expected mean and width.



One issue is that a Gaussian is not limited in range, and has a tail which goes to $\pm\infty$. However, the exponential (and hence sum of exponentials) has a range 0 to ∞ and so it can never truly duplicate the Gaussian distribution for negative x . It is also the case for both of the discrete distributions we have studied, the binomial and the Poisson, that n is limited to $0 \leq n$ (and in the binomial case $n \leq N$). However, in all cases, with a standard deviation much smaller than the mean, then the probabilities near zero are very small and this effect is not important. For the binomial, we would also need the mean to be far from the upper limit N .

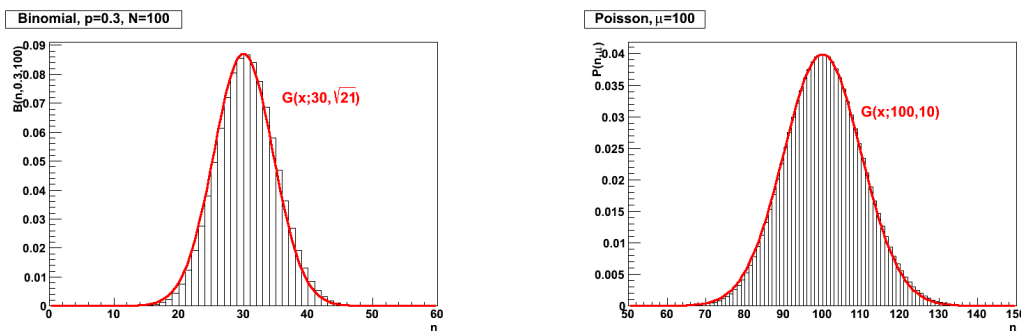
25 CLT for discrete distributions

The CLT applies to discrete distributions but here a bit of care is needed. The sum of the numbers resulting from a binomial or Poisson distribution must be an integer. Indeed, the same is true of the dice case discussed above. Hence, this sum can never be truly a Gaussian distribution as it is still discrete. However, with a large enough sum, then the standard deviation can be much bigger than 1, so the integerisation becomes effectively negligible and the distribution looks almost continuous.

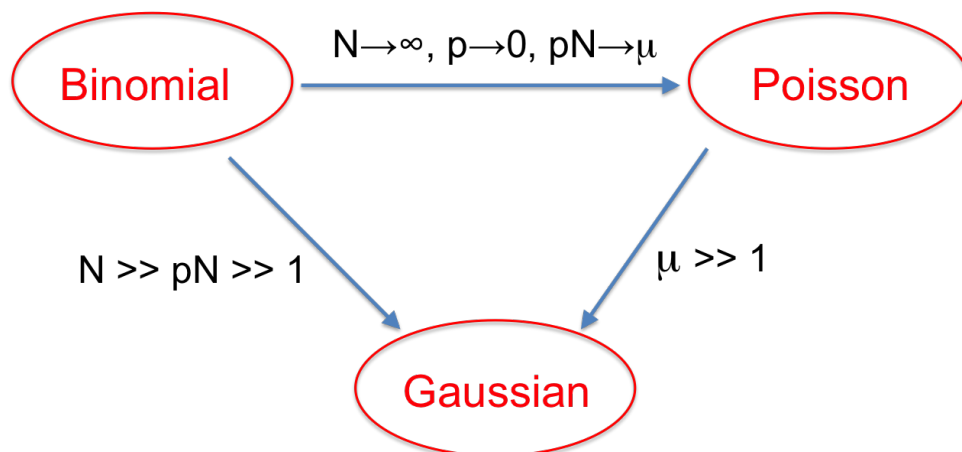
Both the binomial and Poisson are in fact very simple to sum. Specifically, the sum of the outcomes of a binomial is another binomial, and the sum of Poissons is another Poisson. This can be shown mathematically, but in fact, it is straightforward to understand. For the binomial, doing N trials and then another N trials is equivalent to $2N$ trials. This is still binomial,

i.e. the probability distribution of the sum will be given by $B(n; p, 2N)$. For the Poisson, the average number expected is μ and so if the number observed is measured N times, then the total average number expected over these N times is $N\mu$. Summing the n_i observed will be equivalent to having a Poisson with an average $N\mu$ sampled once. Hence, the sum is also a Poisson, explicitly its probability distribution is given by $P(n; N\mu)$.

Therefore, to see how sums of these distributions tend to a Gaussian just requires us to take the number of trials N to be large for the binomial (while ensuring that $1 \ll pN \ll N$ to avoid the problems with discreteness and range), or the average μ to be large for the Poisson. These are shown below for $p = 0.3$ and $N = 100$ for the binomial, and $\mu = 100$ for the Poisson, with the corresponding Gaussian approximations superimposed. It is seen that the Gaussian approximations are very reasonable. The width of the corresponding Gaussian for the Poisson case is seen to be $\sim \sqrt{n}$.



Note, this is a different limit from that discussed in lecture 2 where the Poisson and binomial become the same. The limit there was for a large number of trials, but not necessarily a large mean. The CLT limit of them both becoming Gaussian-like is only when the mean is large (and for the binomial, when the mean is also small compared with N).



26 Non-examinable: Appendix

26.1 Skewness and kurtosis

The explicit forms for expectation value, variance, skewness and kurtosis of a discrete distribution are

$$\begin{aligned} E(x) &= \frac{1}{N} \sum_{i=1}^N x_i P_i \\ V(x) &= \frac{1}{N} \sum_{i=1}^N [x_i - E(x)]^2 P_i \\ S(x) &= \frac{1}{NV(x)^{3/2}} \sum_{i=1}^N [x_i - E(x)]^3 P_i \\ K(x) &= -3 + \frac{1}{NV(x)^2} \sum_{i=1}^N [x_i - E(x)]^4 P_i \end{aligned}$$

with equivalent definitions for continuous distributions. The -3 in the definition of kurtosis is purely to give zero for a Gaussian distribution.

26.2 PDF for the sum of two exponential random variables

The PDF for two exponential random variables is

$$\rho(x_1, x_2) dx_1 dx_2 = \lambda e^{-\lambda x_1} \lambda e^{-\lambda x_2} dx_1 dx_2 = \lambda^2 e^{-\lambda(x_1+x_2)} dx_1 dx_2 = \lambda^2 e^{-\lambda X} dx_1 dx_2$$

To get the PDF in terms of X , we need to change variables from x_1 and x_2 to $X = x_1 + x_2$ and another variable; a simple choice is the difference $D = x_2 - x_1$. Inverting these gives $x_1 = (X - D)/2$ and $x_2 = (X + D)/2$. The Jacobian for this change is

$$\begin{vmatrix} \partial x_1 / \partial X & \partial x_1 / \partial D \\ \partial x_2 / \partial X & \partial x_2 / \partial D \end{vmatrix} = \begin{vmatrix} 1/2 & -1/2 \\ 1/2 & 1/2 \end{vmatrix} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

so that

$$dx_1 dx_2 = \frac{1}{2} dX dD$$

Hence, the PDF in terms of X and D is

$$\rho(X, D) dX dD = \frac{1}{2} \lambda^2 e^{-\lambda X} dX dD$$

We only want the PDF in terms of X (as we don't care about D) so we need to integrate over D for fixed X . The minimum possible value of D is when $x_2 = 0$, for which $x_1 = X$ and hence $D = -X$. Similarly, the maximum possible value of D is when $x_1 = 0$, for which $x_2 = X$ and hence $D = X$. Therefore

$$\rho(X) = \int_{-X}^X \frac{1}{2} \lambda^2 e^{-\lambda X} dD = \frac{1}{2} \lambda^2 e^{-\lambda X} [D]_{-X}^X = \frac{1}{2} \lambda^2 e^{-\lambda X} 2X = \lambda^2 X e^{-\lambda X}$$